

ASSIGNMENT : Python Basics

Question 1 : What is Python and why is it widely used in Data Analytics? Mention any three reasons.

Solution : **Python** is a high-level, easy-to-learn programming language used for many purposes like web development, automation, and data analysis.

Reasons why it is widely used in Data Analytics:

1. **Easy to learn and use** – Simple syntax makes it beginner-friendly.
 2. **Powerful libraries** – Libraries like *Pandas*, *NumPy*, *Matplotlib* help in data analysis.
 3. **Large community support** – Many tutorials and resources are available online.
- Question 2 : Explain the difference between List and Tuple in Python.

Solution :

Feature	List	Tuple
Mutability	Mutable (can change)	Immutable (cannot change)
Syntax	[]	()
Performance	Slower	Faster
Example	[1, 2, 3]	(1, 2, 3)

Question 3 : What is a function in Python? Why are functions useful?

Solution : A **function** is a block of code that performs a specific task.

Why functions are useful:

- Avoid repetition of code
- Make programs organized
- Easy to reuse and maintain

Question 4 : Write a Python program to take a user's name as input and print a greeting message.

Solution :

```
name = input("Enter your name: ")  
  
print("Hello,", name, "! Welcome!")
```

Question 5 : Write a Python program to check whether a number is even or odd using conditional statements.

Solution :

```
num = int(input("Enter a number: "))
```

```
if num % 2 == 0:
```

```
    print("Even number")
```

```
else:
```

```
    print("Odd number")
```

Question 6 : Write a program to print numbers from 1 to 10 using a loop.

Solution :

```
for i in range(1, 11):
```

```
    print(i)
```

Question 7 : Create a list of five numbers and print the maximum number from the list.

Solution :

```
numbers = [10, 25, 5, 40, 15]
```

```
print("Maximum number is:", max(numbers))
```

Question 8 : Write a Python program to remove duplicate values from a list using a set.

Solution :

```
numbers = [1, 2, 2, 3, 4, 4, 5]
```

```
unique_numbers = list(set(numbers))
```

```
print(unique_numbers)
```

Question 9 : Write a function that returns the square of a number.

Solution :

```
def square(num):
```

```
    return num * num
```

```
print(square(5))
```

Question 10 : Write a Python program to count how many times a number appears in a list. Example List [2,3,4,2,5,2]

Solution :

```
numbers = [2, 3, 4, 2, 5, 2]
```

```
count = numbers.count(2)
```

```
print("2 appears", count, "times")
```